Ben Efron, PhD | Systems Neuroscientist | Neural Computation & Bio-Inspired System Design

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PROFESSIONAL SUMMARY

Experimental systems neuroscientist studying how neural circuits compute and adapt, and translating these principles into adaptive sensing, closed-loop control, and system design. **7+ years** leading experiments and multimodal data pipelines; collaborate closely with engineers to bridge biological computation and technology. Currently Postdoctoral Researcher at the University of Liège's Brain-Inspired Computing Lab in collaboration with imec.

CORE STRENGTHS

Systems Neuroscience

- Neural population dynamics & sensory coding
- Synaptic plasticity modeling
- Adaptive computation & feedback mechanisms
- Translating biological design to engineered systems

Data Analysis & Modeling

- Data analysis & neural decoding
- Statistical modeling (GLM, Bayesian, regression)
- Signal processing & feature extraction
- Machine learning & neural data science

Experimental Design & Automation

- Experimental automation
- Hypothesis testing & controls design
- Electrophysiology & multi-sensor acquisition
- Rapid prototyping (CAD, Arduino, Bonsai)

TECHNICAL EXPERTISE

Systems Neuroscience ◦ **Neuromorphic Engineering** ◦ **Neuroengineering**
Computational Neuroscience ◦ Signal Processing ◦ Statistical Modeling ◦ Data Analysis
Applied Machine Learning ◦ Python ◦ MATLAB ◦ Experimental Design ◦ Scientific Communication

EXPERIENCE

Brain-Inspired Computing Lab, University of Liège
Postdoctoral Researcher *2024–Present*

- Coordinate with imec engineers to refine high-density recording and closed-loop experimentation; mentor students/interns.
- Study population dynamics & adaptive circuit computation; develop analysis pipelines linking activity patterns to algorithmic design.
- Provide biological insight for modeling synaptic plasticity and adaptive control.

Weizmann Institute of Science

PhD & MSc Researcher *2017–2024*

- Investigated detection and neural encoding of whisker-generated sounds in behaving mice.
- Built multimodal electrophysiology + behavioral systems (neural, audio, motion) with synchronized acquisition and automation.
- Applied ML/statistical models to decode neural–behavior relationships; presented at international meetings.
- Teaching Assistant, *Synaptic Physiology Course* — membrane and cable equations, H&H model, RC circuits, Nernst potentials, and synaptic release modeling.

Hebrew University (Soreq Lab)

Undergraduate Researcher *2014–2016*

- Studied microRNA-based regulation of neural signaling using bioinformatics and molecular assays.

EDUCATION

PhD + MSc: Systems Neuroscience — Weizmann Institute of Science

BSc: Psychobiology (Cum Laude) — Hebrew University of Jerusalem

Exchange: University of Melbourne

SELECTED PUBLICATIONS

2025: **Efron, B.**, Ntelezos, A., Katz, Y., & Lampl, I. *Neural encoding and behavioral detection of whisker-generated sounds in mice. Current Biology.* Article link

2023: **Efron, B.**, Katz, Y., & Lampl, I. Object-specific neuronal response during active whisking. *Batsheva Conference* — Selected short talk.

2020: **Efron, B.**, Katz, Y., & Lampl, I. Auditory responses to whisker-generated tactile sounds. *BARRELS XXXIII* — Selected short talk.

2018: Hanin, G., **Efron, B.**, et al. miRNA-132 induces hepatic steatosis and hyperlipidaemia. *Gut*, 67(6):1124–1134.

HONORS & SCHOLARSHIPS

WBI Postdoctoral Excellence Scholarship ∘ Schulich Leader STEM Scholarship

Best Poster Award ∘ Magna Cum Laude

LANGUAGES

English — Full Professional ∘ Hebrew — Native